Shreshtha Garg

Full Stack Developer

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Education

ABV-Indian Institute of Information Technology and Management, Gwalior 2022 - 2027 (Expected)
Integrated Post Graduate M.Tech, Information Technology CGPA: 9.00
SDS Convent School, India 2020 - 2021
12th, Non-Medical (CBSE) Percentage: 96.4%

Personal Projects

ClickON - AI Quick-Commerce Engine (Amazon Hackon) | Live Demo | GitHub June 2026
Node.js, Express, JavaScript, MongoDB, Gemini API

- Engineered an intent-driven quick-commerce engine to eliminate shopping friction, processing **3 input types** (text, voice, image) via Gemini API to instantly translate abstract user goals into purchasable carts.
- Developed **2 distinct checkout flows** ("Flash" and "Quick" modes) to reduce decision fatigue, dynamically assembling **3 deterministic product tiers** (Budget, Balanced, Premium) using an $O(K \log K)$ value-sorting algorithm.
- Architected a zero-input predictive discovery feed that proactively curates personalized smart bundles by analyzing real-time weather, time of day, purchase history, and hyper-local community order trends.
- Optimized backend product retrieval by implementing **MongoDB ESR (Equality, Sort, Range) indexing** for $O(\log N)$ lookups, strictly decoupling the transactional cart logic from the non-deterministic LLM pipeline.

HostelMate - Hostel Grievance Portal | Live Demo | GitHub Nov 2024
MERN Stack, Tailwind CSS, Redis, Socket.io

- Developed a **real-time** grievance portal with **Socket.io** to support **instantaneous chat** and status updates across **3 primary user roles** (Student, Warden, Admin).
- Built a secure, **role-based authentication** system for **3 distinct user types**, protecting over **12 different API routes** based on user permissions.
- Boosted login performance by implementing **OTP verification with Redis**, eliminating database queries during token validation and reducing login latency.
- Integrated email notifications with **Nodemailer** and a responsive UI using **React.js** and **Tailwind CSS**.

Maximal Concurrent Rare-Node Activation for Hardware Trojan Detection July 2026
C++, ATPG, DFS, Backtracking, ISCAS85

- Developed a **multi-threshold ATPG framework** for concurrent rare-node activation in Hardware Trojan detection.
- Designed a **4 stage ATPG pipeline** for multi-threshold sweeping, pattern compaction, and targeted deep search.
- Achieved **321 concurrent rare-node activations** on **c3540**, cutting test patterns by **93.7%** (**2744→172**) and scaling trigger combinations $\mathbf{1.3 \times 10^{28} \rightarrow 5.11 \times 10^{96}}$.

Experience

Firstbench.ai Remote
Software Developer Intern Aug 2024 - Feb 2025

- Developed and shipped **4 new features** for a customer-facing web application, creating over **20 reusable React components** and **7 new RESTful API endpoints**.
- Collaborated within a **cross-functional agile team** comprising frontend, backend, UI/UX, and AI specialists alongside a product manager to deliver features in **2-week sprint cycles**.
- Optimized backend performance by **refactoring database queries**, reducing the average **API response time** for the main dashboard by an estimated **20%**.

Technical Skills

- **Languages**: C++, Java, JavaScript, Python, HTML, CSS, SQL
- **Frameworks & Libraries**: React.js, Next.js, Node.js, Express.js, Tailwind CSS
- **Databases**: MongoDB, MySQL, PostgreSQL
- **Tools & Technologies**: Generative AI, AWS, Git, Docker, RESTful APIs, JWT Authentication, Redis, Automation
- **Core Competencies**: Software Development, System Architecture, Algorithms, Analytics, Agile Methodologies

Competitive Programming

- **LeetCode Rating:** 2048 (Knight) - 1000+ Problems Solved | leetcode.com/u/ShreshthaGarg037/
- **Codeforces Rating:** 1640 (Expert) - 400+ Problems Solved | codeforces.com/profile/ShreshthaGarg037
- **CodeChef Rating:** 1866 (4 Star) - codechef.com/users/raghavgarg037

Achievements

- Secured **Institute Rank 5** in **ICPC 2024 Qualifiers** 2024
- Achieved rank **318** in LeetCode Weekly 495, rank **657** in LeetCode Weekly 462, rank **677** in Biweekly 179, and rank **805** in Codeforces Round 1093 and rank **993** in Round 1118 2025-2026
- Secured **AIR 8707** in **JEE Advanced** & **AIR 10494** in **JEE Mains** 2022
- Achieved **District Rank 7** in **CBSE 10th Board Examinations** 2019